

KETTSTAKA, SwedenWMO: **025540**Lat: **58.7162N**Lon: **15.0271E**Elev: **738**StdP: **14.31**Time Zone: **1.00 (EUC)**Period: **01-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.7	11.5	2.6	6.5	10.6	6.9	8.0	14.5	20.8	32.4	18.9	32.6	4.6	350	0.588

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.8	77.8	62.9	74.6	61.5	71.5	60.1	65.6	73.0	63.8	70.8	62.1	68.5	6.2	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.0	88.6	67.7	61.3	83.2	66.6	59.3	77.4	65.1	30.8	73.3	29.5	71.0	28.2	68.6	73.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.5	14.6	13.3	4.8	81.5	6.4	3.0	0.2	83.7	-3.5	85.5	-7.1	87.1	-11.7	89.3		
(5)			4.2	68.3	5.9	2.3	0.0	69.9	-3.4	71.3	-6.7	72.5	-11.0	74.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.0	27.0	27.6	33.3	42.4	51.1	58.0	62.5	60.6	52.7	43.9	36.3	31.1
	DBStd	14.01	7.71	7.59	7.23	6.37	6.54	5.52	5.48	4.61	4.98	6.18	6.74	8.33
	HDD50	3395	713	626	519	241	62	3	0	1	29	203	412	586
	HDD65	7745	1178	1046	984	678	431	221	118	151	370	655	862	1051
	CDD50	1196	0	0	0	13	97	244	389	331	109	13	0	0
	CDD65	70	0	0	0	0	1	12	41	16	0	0	0	0
	CDH74	330	0	0	0	0	10	68	204	48	0	0	0	0
	CDH80	28	0	0	0	0	0	2	21	5	0	0	0	0
Wind	WSAvg	7.0	7.4	7.1	7.4	7.0	6.7	6.4	6.0	6.1	6.7	7.3	8.0	7.7
Precipitation	PrecAvg	25.2	1.7	1.5	1.2	1.5	1.8	2.6	3.3	2.8	2.3	2.3	2.3	1.9
	PrecMax	32.6	3.1	2.8	2.0	3.6	3.2	4.9	7.3	6.7	5.7	5.1	4.6	4.1
	PrecMin	19.9	0.6	0.4	0.2	0.3	0.6	0.1	0.9	0.6	0.5	0.9	0.7	0.5
	PrecStd	3.1	0.8	0.6	0.4	0.9	0.8	1.2	1.7	1.4	1.4	1.1	1.0	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.3	44.7	54.9	64.8	75.4	79.6	82.7	78.5	69.2	59.8	51.7	48.2
		MCWB	40.8	41.4	45.7	50.8	59.0	62.0	65.0	64.1	59.7	55.2	49.9	46.8
	2%	DB	39.9	41.8	49.4	61.0	71.0	75.7	79.2	74.7	65.3	55.7	49.0	45.6
		MCWB	38.6	39.4	42.0	49.2	56.5	60.6	63.7	63.1	57.9	52.8	47.5	44.2
	5%	DB	38.3	39.7	45.4	56.6	67.0	72.0	75.9	71.3	62.2	53.9	47.1	43.1
		MCWB	37.0	37.8	40.1	46.7	54.8	58.7	62.5	61.4	56.5	51.5	45.7	41.6
	10%	DB	36.6	37.6	42.4	52.9	62.7	68.3	72.4	68.3	59.7	52.1	45.0	41.2
		MCWB	35.7	36.1	38.3	44.3	52.2	56.8	61.3	60.1	55.0	50.1	43.9	40.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.3	42.8	46.0	53.6	61.8	64.7	68.3	67.6	61.2	55.7	50.3	47.1
		MCDB	41.9	43.9	53.1	63.0	71.3	73.8	77.0	73.6	66.5	57.7	50.9	48.0
	2%	WB	39.0	40.2	42.9	49.9	59.1	62.3	66.2	65.2	58.8	53.8	47.8	44.3
		MCDB	39.6	41.1	47.3	57.9	67.0	72.3	74.5	71.2	63.3	55.3	48.6	45.3
	5%	WB	37.3	38.0	41.0	47.7	56.3	60.1	64.6	63.2	57.3	52.1	45.9	41.9
		MCDB	38.0	39.4	44.6	54.9	63.8	69.7	72.3	68.2	61.2	53.5	46.8	42.8
	10%	WB	35.7	36.2	39.1	45.5	53.7	58.2	62.9	61.6	55.7	50.4	44.0	39.9
		MCDB	36.5	37.6	41.8	51.7	61.2	66.2	69.9	66.5	59.0	51.9	44.9	41.0
Mean Daily Temperature Range	5% DB	MDBR	6.4	7.2	9.9	13.5	14.5	13.9	12.8	11.5	10.1	8.1	6.4	6.2
		MCDBR	7.6	7.8	13.8	18.6	19.5	18.6	17.5	15.5	13.2	8.9	7.8	7.0
		MCWBR	7.5	6.5	9.0	10.6	10.8	10.4	8.4	8.3	7.3	6.8	7.4	7.1
	5% WB	MCDBR	7.6	7.3	12.7	16.8	17.6	16.8	15.6	13.4	12.1	8.5	7.4	7.0
		MCWBR	7.6	6.6	8.7	10.4	10.9	10.2	8.5	7.9	7.4	7.3	7.2	7.1
Clear-Sky Solar Irradiance	taub		0.270	0.308	0.329	0.366	0.362	0.351	0.374	0.367	0.337	0.327	0.295	0.254
	taud		2.228	2.302	2.373	2.339	2.410	2.461	2.413	2.432	2.503	2.443	2.295	2.157
	Ebn at Noon		178	227	259	266	276	281	270	264	256	220	166	139
	Edh at Noon		14	22	27	33	33	32	33	30	24	19	14	11
All-Sky Solar Radiation	RadAvg		108	302	785	1259	1622	1820	1632	1293	873	395	138	74
	RadStd		16	51	89	142	144	93	191	138	118	66	27	7